

Unit 110 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 62.0 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 315 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	203 sq ft/ac (SE 3.7 %)	Type	Sierran mixed conifer
Trees per acre	137	CWHR size / density	4 / D (cover 82 %)
QMD	16.5 in	SDI	287 (50 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	194 sq ft/ac 18.4 in	BA QMD	10 sq ft/ac 7.5 in
TPA	105	TPA	32
Volume	34.1 MBF/ac (6200 cu ft)	Volume	4.5 green tons/ac (150 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~142 sq ft/ac BA	Sampling error	7.5 % at 95 % (t = 2.04, n = 31)
Species (BA)	PP 65, WF 60, IC 28, DF 26, SP 25	Stratum standard	40 % - MEETS; 31 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	6	18.9	3.9	50	0.3
8-11.9	27	31.5	17.4	340	1.9
12-15.9	60	36.7	38.7	916	5.0
16-19.9	68	25.4	43.9	1255	6.9
20-23.9	46	11.5	29.7	888	4.9
24-27.9	30	5.4	19.4	701	3.9
28-31.9	32	4.3	20.6	798	4.4
32-35.9	15	1.6	9.7	396	2.2
36-39.9	10	0.8	6.5	291	1.6
40-43.9	5	0.3	3.2	156	0.9
44-47.9	5	0.3	3.2	169	0.9
48-51.9	3	0.1	1.9	108	0.6
52-55.9	3	0.1	1.9	106	0.6
56-59.9	2	0.1	1.3	68	0.4
60-63.9	3	0.1	1.9	108	0.6
All	315	137.0	203.2	6350	34.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	101	46.5	65.2	16.0	1987	10.9
WF	93	36.5	60.0	17.4	1893	10.4
IC	43	21.4	27.7	15.4	860	4.7
DF	40	16.1	25.8	17.1	855	4.7
SP	38	16.5	24.5	16.5	755	4.2
All	315	137.0	203.2	16.5	6350	34.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 110: plots on the 300 ft grid (orange = QA flag)

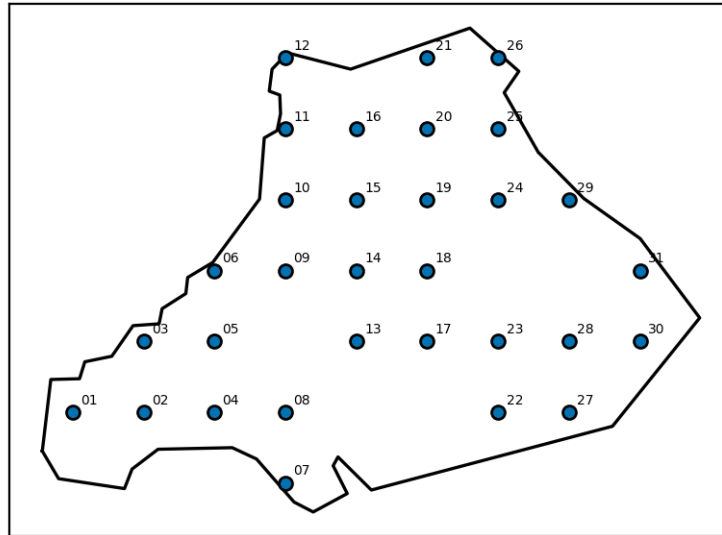


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
110-01	6	120	68	ok
110-02	13	260	89	ok
110-03	11	220	50	ok
110-04	8	160	54	ok
110-05	7	140	25	ok
110-06	10	200	82	ok
110-07	13	260	131	ok
110-08	13	260	85	ok
110-09	12	240	57	ok
110-10	6	120	114	ok
110-11	10	200	141	ok
110-12	7	140	74	ok
110-13	9	180	117	ok
110-14	11	220	98	ok
110-15	10	200	83	ok
110-16	11	220	145	ok
110-17	12	240	76	ok
110-18	10	200	121	ok
110-19	10	200	55	ok
110-20	12	240	116	ok
110-21	13	260	30	ok
110-22	11	220	37	ok
110-23	11	220	48	ok
110-24	10	200	85	ok
110-25	10	200	59	ok
110-26	8	160	81	ok
110-27	12	240	110	ok
110-28	10	200	61	ok
110-29	12	240	80	ok
110-30	7	140	170	ok
110-31	10	200	87	ok